

DIGITAL CITY DESIGN WORKSHOP

Final Presentation

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senseable
city lab.



How can a **green corridor map** for Gangnam be developed using **street-view images** and Green View Index (**GVI**) analysis?

- How are **GVI values spatially distributed** across Gangnam District?
- **Which roads and street networks** are most **effective** for establishing continuous green corridors in Gangnam District?
- What **planning and policy strategies** can support the implementation of the proposed green corridor network?

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Importance of Green Corridor

Urban biodiversity problem

- Habitat loss
- Habitat fragmentation
- Decline in biodiversity



Landscape connectivity

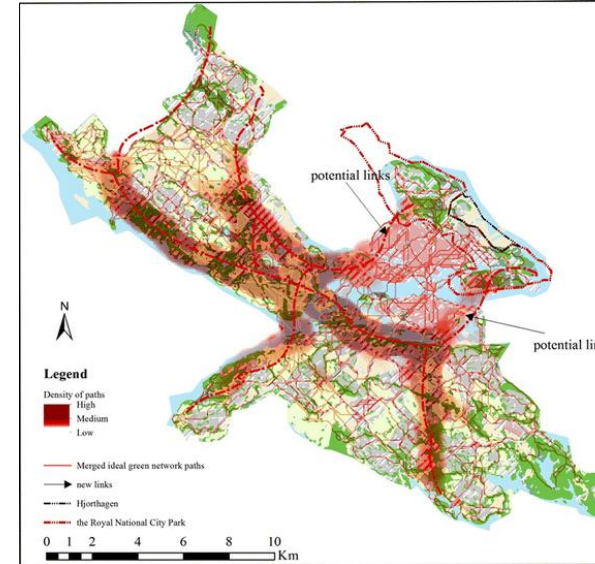
by

Green Corridor



Amsterdam

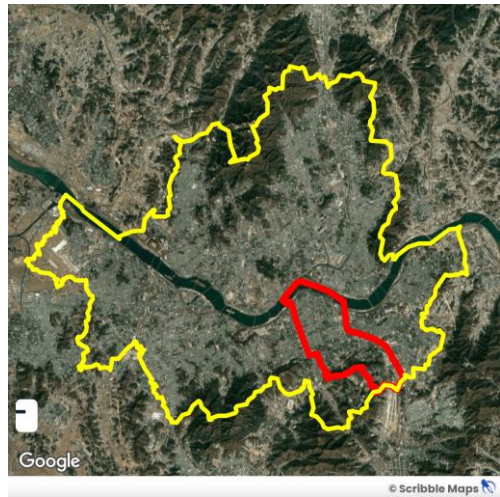
City of Amsterdam. (2020). Amsterdam green infrastructure vision 2050: A liveable city for people, plants, and animals. Department of Planning and Sustainability, City of Amsterdam.
https://carbonneutralcities.org/wp-content/uploads/2020/09/Amsterdam-Green-Infrastructure-Vision-2050_toegankelijk_02092020.pdf



Stockholm

Xiu, N., Ignatieva, M., van den Bosch, C. K., Chai, Y., Wang, F., Cui, T., & Yang, F. (2017). A socio-ecological perspective of urban green networks: the Stockholm case. *Urban ecosystems*, 20(4), 729-742.

Characteristic of Gangnam



Han River Park (Visit Gangnam)

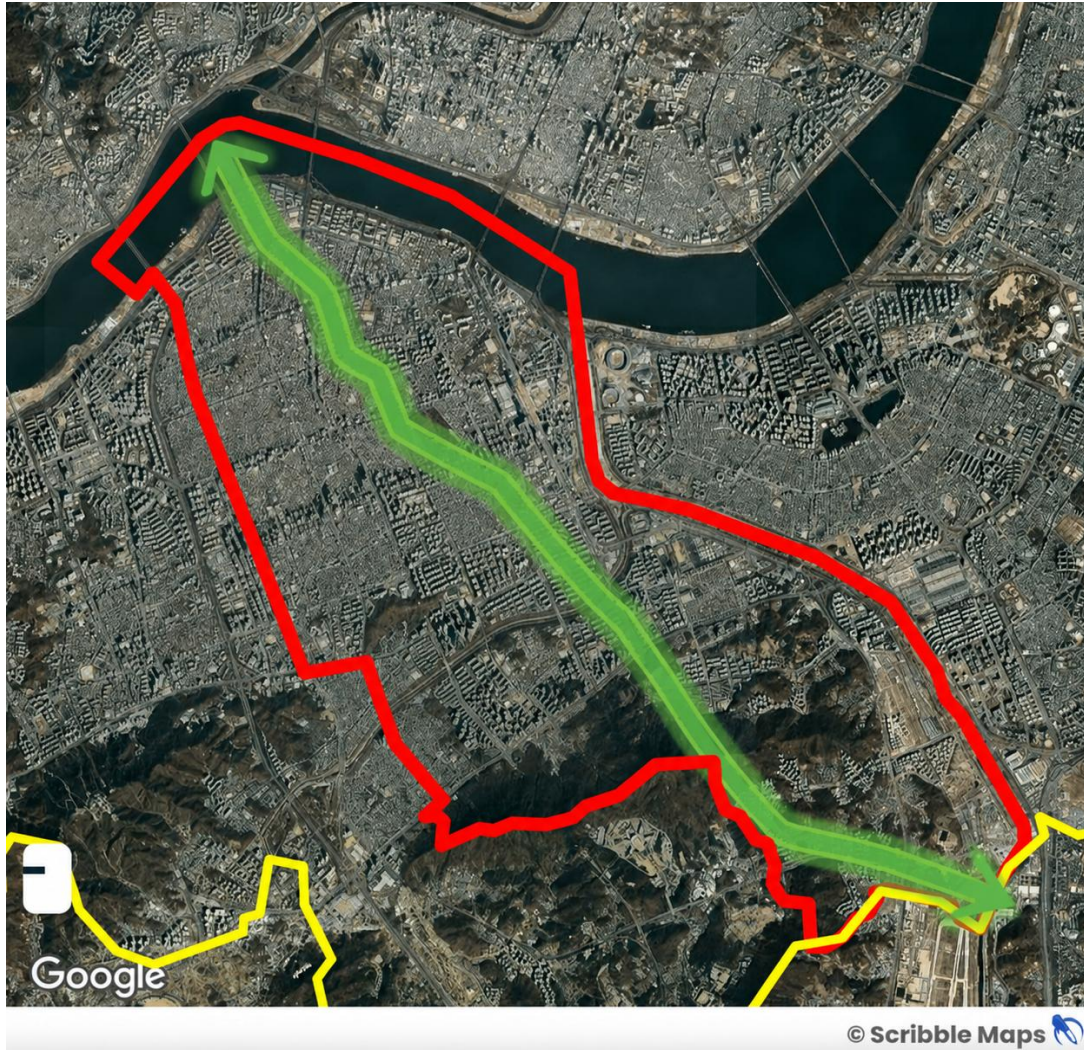


Business district (Visit Gangnam)



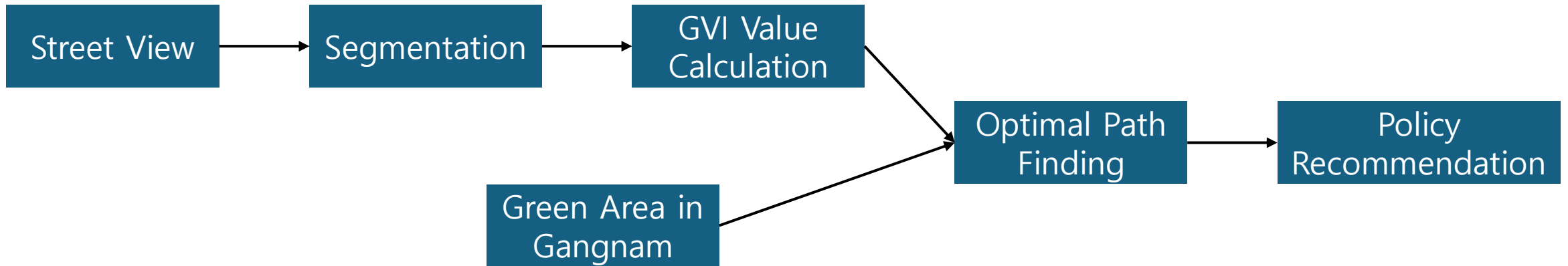
Daemo Mountain (Visit Gangnam)

Research Question



What is the most **effective** way to **connect green spaces in the Gangnam** by **using street images**?

Pipeline



1. Street View



Identifying roads using OpenStreetMap



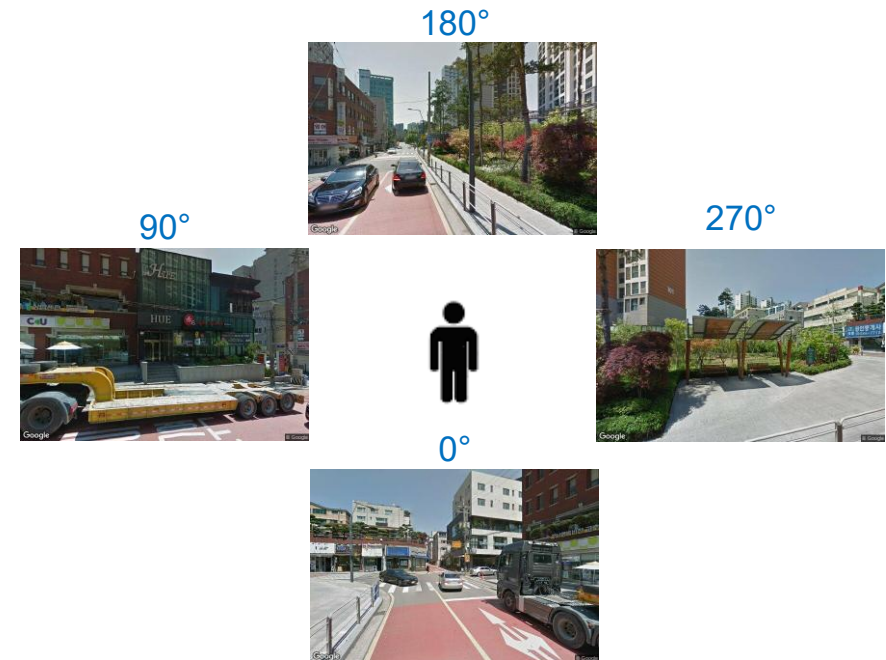
Place a marker and return coordinates every **25 meter** (20,037 points)



Download the Google Street View images for the location closest to each coordinate



Discard point with error for download the data (2,361 points)



2. Segmentation

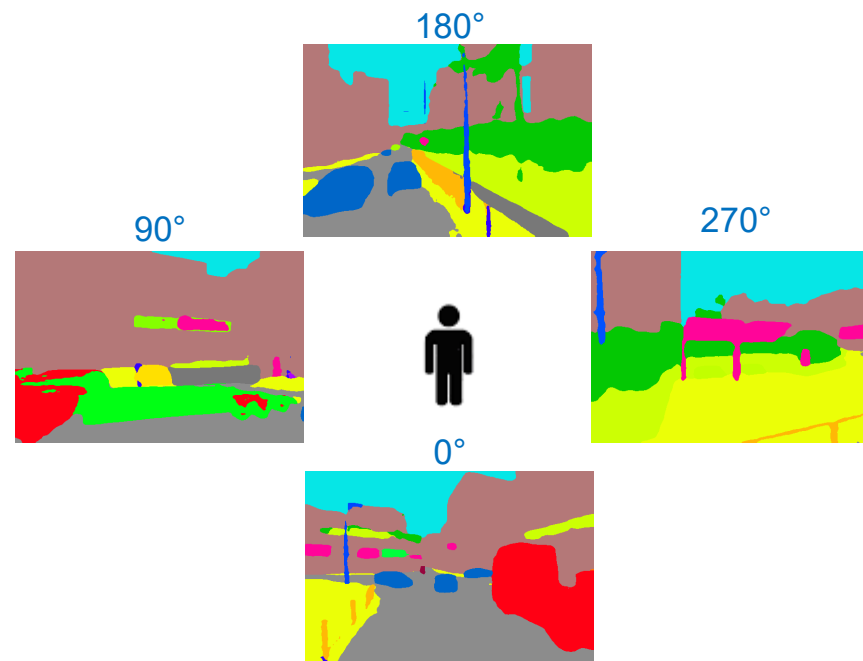
Semantic Segmentation using Mask2Former

- Developed by Meta AI
- Capable of distinguishing 150 semantic classes, including trees and grass



Pixel-wise proportion of each semantic category
extracted from every image

panoid	lat	lon	filename	year	wall	building	sky	floor	tree	ceiling	road	bed	window	grass
P813VotorUj856--i_yKxg	37.50998	127.0398	P813Votoi	2018-05	0.03	0.34	0.15	0	0.03	0	0.27	0	0	0
P813VotorUj856--i_yKxg	37.50998	127.0398	P813Votoi	2018-05	0.04	0.44	0.08	0	0	0	0.19	0	0	0
P813VotorUj856--i_yKxg	37.50998	127.0398	P813Votoi	2018-05	0	0.28	0.1	0	0.13	0	0.18	0	0	0
P813VotorUj856--i_yKxg	37.50998	127.0398	P813Votoi	2018-05	0	0.22	0.11	0	0.23	0	0.09	0	0	0.03
HAwK mz5AbekXOt_oTyKleQ	37.5101	127.0397	HAwK mz5	2018-05	0	0.33	0.12	0	0.01	0	0.26	0	0	0
HAwK mz5AbekXOt_oTyKleQ	37.5101	127.0397	HAwK mz5	2018-05	0.03	0.56	0.03	0	0	0	0.13	0	0	0
HAwK mz5AbekXOt_oTyKleQ	37.5101	127.0397	HAwK mz5	2018-05	0.03	0.32	0.12	0	0.15	0	0.12	0	0	0
HAwK mz5AbekXOt_oTyKleQ	37.5101	127.0397	HAwK mz5	2018-05	0	0.21	0.18	0	0.15	0	0	0	0	0
oXcmsD4FWseKjrhD2N7CPQ	37.50998	127.0399	oXcmsD4F	2018-05	0.11	0.47	0.06	0	0.02	0	0.02	0	0	0
oXcmsD4FWseKjrhD2N7CPQ	37.50998	127.0399	oXcmsD4F	2018-05	0	0.55	0.11	0	0	0	0.17	0	0	0
oXcmsD4FWseKjrhD2N7CPQ	37.50998	127.0399	oXcmsD4F	2018-05	0	0.57	0.01	0	0.06	0	0.13	0	0	0.01
oXcmsD4FWseKjrhD2N7CPQ	37.50998	127.0399	oXcmsD4F	2018-05	0	0.24	0.08	0	0.18	0	0.34	0	0	0
4JKkNtKxR_qSHcB1So7AyA	37.51004	127.0401	4JKkNtKxR	2018-05	0	0.86	0	0	0	0	0.05	0	0	0
4JKkNtKxR_qSHcB1So7AyA	37.51004	127.0401	4JKkNtKxR	2018-05	0	0.47	0.12	0	0	0	0.21	0	0	0
4JKkNtKxR_qSHcB1So7AyA	37.51004	127.0401	4JKkNtKxR	2018-05	0.01	0.66	0	0	0.07	0	0.1	0	0	0
4JKkNtKxR_qSHcB1So7AyA	37.51004	127.0401	4JKkNtKxR	2018-05	0.02	0.63	0.01	0	0.03	0	0.24	0	0	0
JDW7JEmXqCEbKhFGt_QO9A	37.51013	127.0404	JDW7JEm	2018-05	0.09	0.36	0.14	0	0.08	0	0.22	0	0	0
JDW7JEmXqCEbKhFGt_QO9A	37.51013	127.0404	JDW7JEm	2018-05	0.06	0.42	0.14	0	0.03	0	0.16	0	0	0
JDW7JEmXqCEbKhFGt_QO9A	37.51013	127.0404	JDW7JEm	2018-05	0	0.56	0	0	0.01	0	0.22	0	0	0
JDW7JEmXqCEbKhFGt_QO9A	37.51013	127.0404	JDW7JEm	2018-05	0.02	0.57	0	0	0.01	0	0.29	0	0	0
zMcxS6nHmXAuiXEZUoptqW	37.50977	127.0399	zMcxS6nH	2018-05	0	0.32	0.15	0	0.04	0	0.33	0	0	0
zMcxS6nHmXAuiXEZUoptqW	37.50977	127.0399	zMcxS6nH	2018-05	0	0.6	0	0	0	0	0.14	0	0	0
zMcxS6nHmXAuiXEZUoptqW	37.50977	127.0399	zMcxS6nH	2018-05	0	0.48	0.09	0	0.03	0	0.23	0	0	0
zMcxS6nHmXAuiXEZUoptqW	37.50977	127.0399	zMcxS6nH	2018-05	0	0.45	0.04	0	0.1	0	0	0	0	0
SWgdGbwzuuS25qlujhdcw	37.50958	127.0399	SWgdGbw	2018-05	0.01	0.37	0.13	0	0.04	0	0.35	0	0	0
SWgdGbwzuuS25qlujhdcw	37.50958	127.0399	SWgdGbw	2018-05	0.01	0.47	0.07	0	0.02	0	0.19	0	0	0
SWgdGbwzuuS25qlujhdcw	37.50958	127.0399	SWgdGbw	2018-05	0.1	0.24	0.1	0	0.19	0	0.22	0	0	0
SWgdGbwzuuS25qlujhdcw	37.50958	127.0399	SWgdGbw	2018-05	0.19	0.36	0	0	0.18	0	0.05	0	0	0
Jpv698SwYakra-3sgpTjtA	37.50938	127.0401	Jpv698Sw	2018-05	0.01	0.34	0.12	0	0.12	0	0.17	0	0	0
Jpv698SwYakra-3sgpTjtA	37.50938	127.0401	Jpv698Sw	2018-05	0	0.48	0	0	0.21	0	0.11	0	0	0.02
Jpv698SwYakra-3sgpTitA	37.50938	127.0401	Jpv698Sw	2018-05	0	0.36	0.12	0	0.08	0	0.35	0	0	0



3. Calculate GVI Value

GVI (Green View Index) Value: Percentage of visible greenery at street level

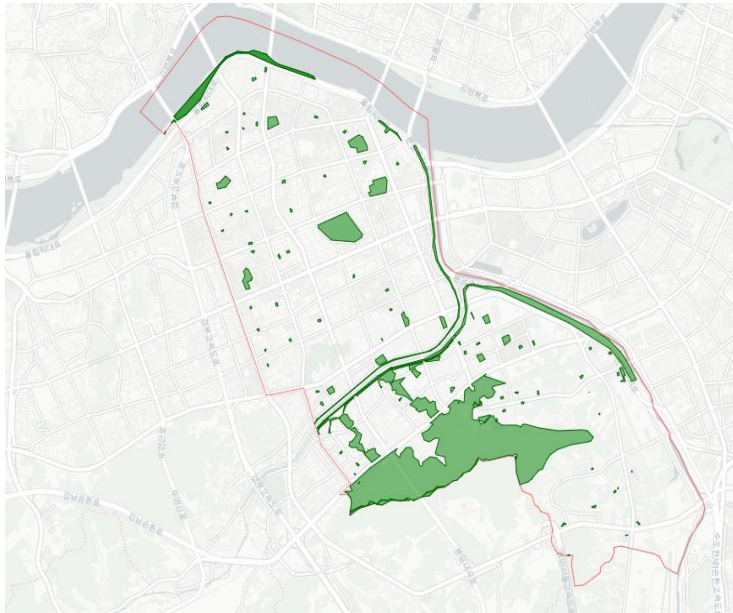
$$P_{gvi} = P_{tree} + P_{grass} + P_{flower} + P_{plant} + P_{palmtree}$$

(averaged by each point – 4 pictures)

GVI value classification	
Very low vegetation	< 5.0%
Low vegetation	5.0% ~ 15.0%
Middle vegetation	15.0% ~ 25.0%
High vegetation	25.0% ~ 35.0%
Very high vegetation	> 35.0%

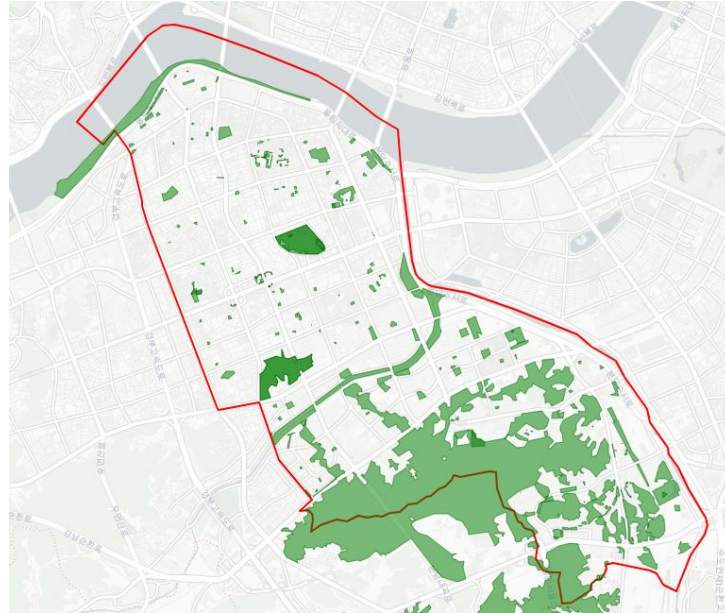
Ma, Y. (2020). Analysis of green view index in Beijing Central Axis Road system based on street view image. *J. Chin. Urban For.*, 18(03), 46-50.

4. Green Area in Gangnam



Park Spatial Information Data

Seoul Open Data Hub
(<https://data.seoul.go.kr/dataList/OA-15529/S/1/datasetView.do>)



Wood & Grassland Data

Open street map



Threshold

$300m^2$

(Too small for bio-diversity)

5. Optimal Path finding

Merge overlapping polygons into a single polygon



Assign representative points



Construct a road network along streets



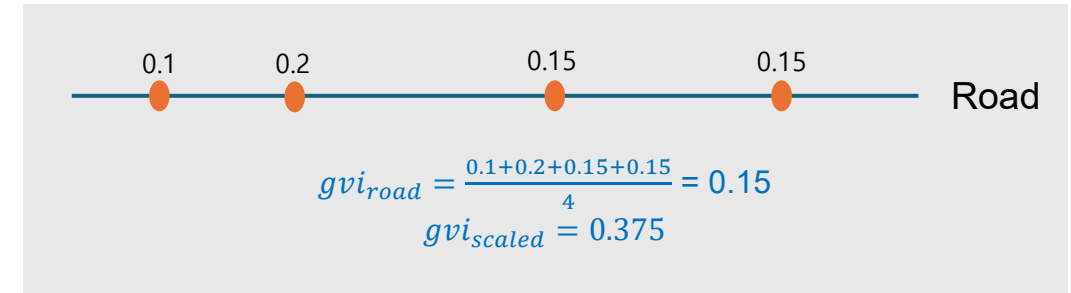
Assign average GVI to each road segment



$$C = L \times (1 + \delta \times (1 - g_{scaled}))$$



Generate a network connecting all green areas with minimum cost



C: Cost

L: Length of the Road

δ : Weight (0, 4, 16, 64)

$$g_{scaled} = \max(0, \min\left(\frac{gvi}{0.4}, 1\right))$$

6. Policy Recommendation

Identify areas in the city with insufficient GVI and further increase urban greenery.



After assigning region-specific weights, **increase the GVI value** to above 0.35 in areas **along the optimal route**.



Adapted from London urban greening case studies

Result 1 – Overall GVI of Gangnam

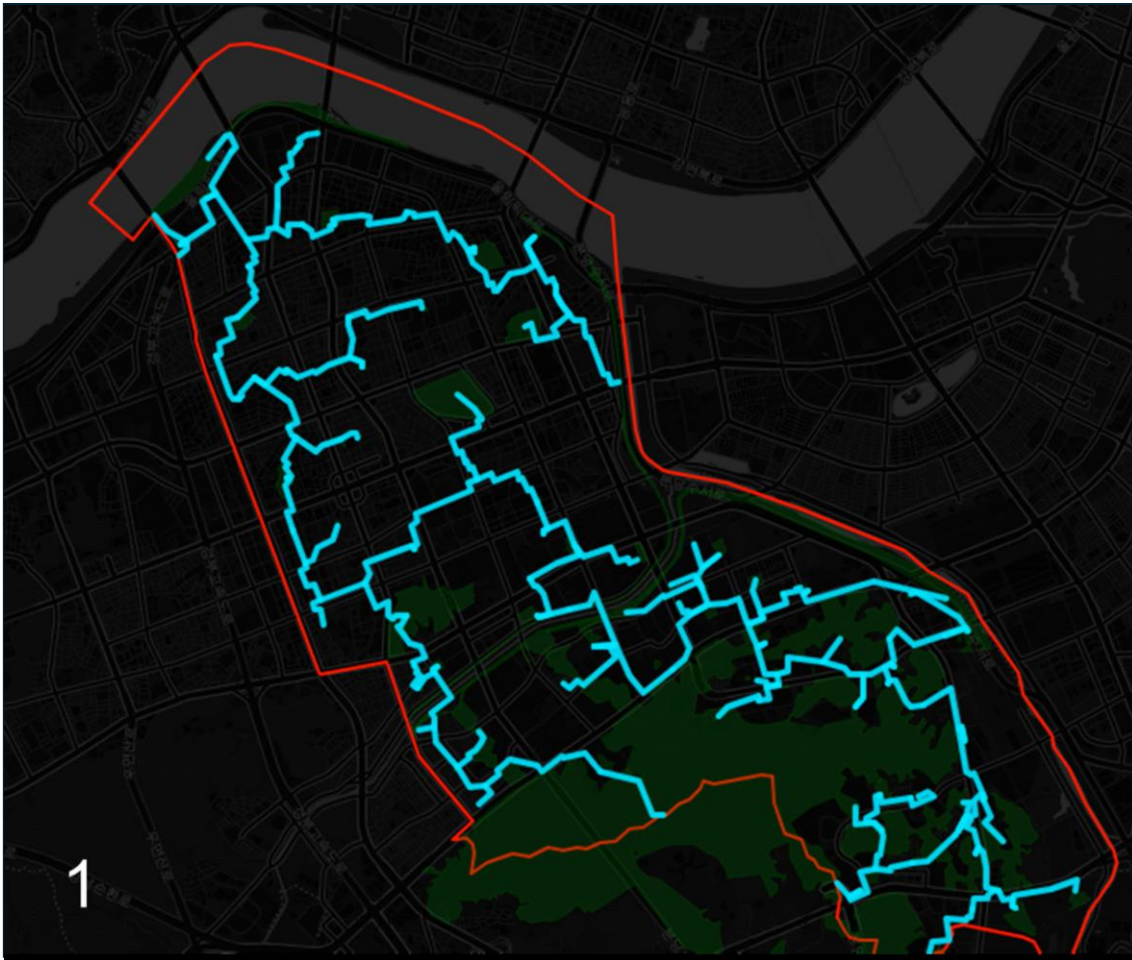


GVI value classification

Very low vegetation	28% (4963)
Low vegetation	31% (5409)
Middle vegetation	20% (3481)
High vegetation	12% (2109)
Very high vegetation	10% (1714)

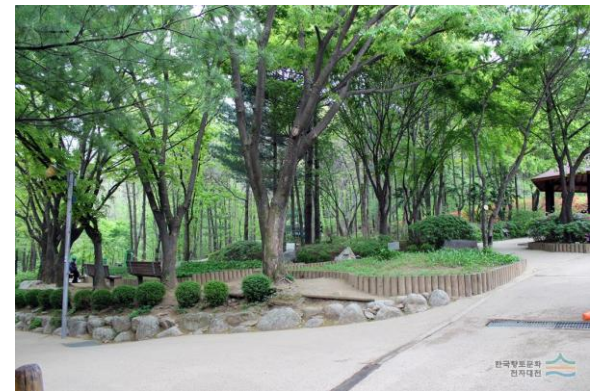
Ma, Y. (2020). Analysis of green view index in Beijing Central Axis Road system based on street view image. *J. Chin. Urban For.*, 18(03), 46-50.

Result 2 – Optimal path finding

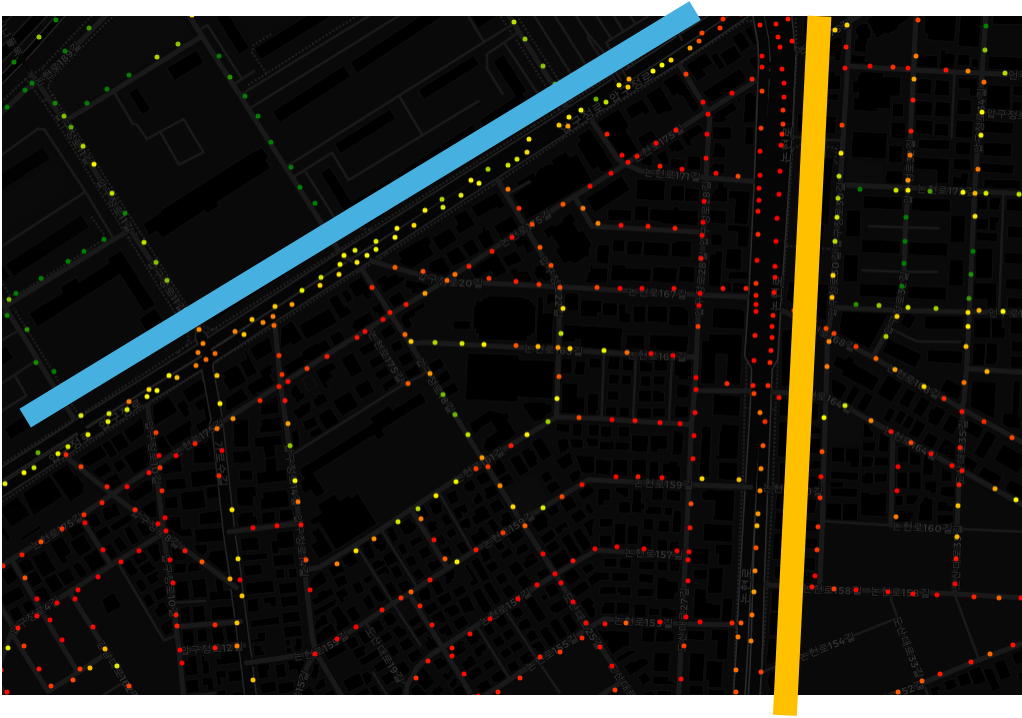


Path	Weight	Length	Average GVI
1	0	66.1 km	0.140
2	4	69.1km	0.166
3	16	74.6 km	0.173
4	64	76.5 km	0.180

Result 3 – Example of Policy Recommendation

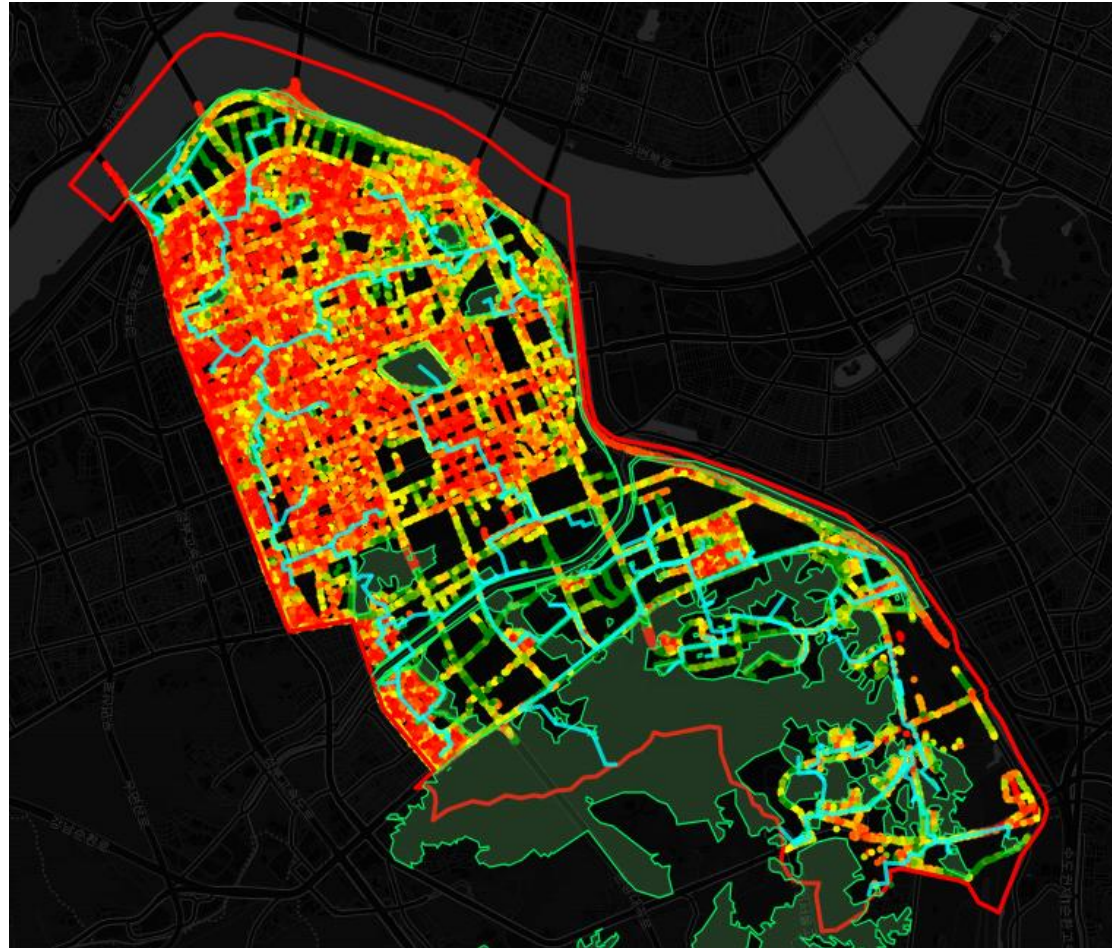


Result 3 – Example of Policy Recommendation

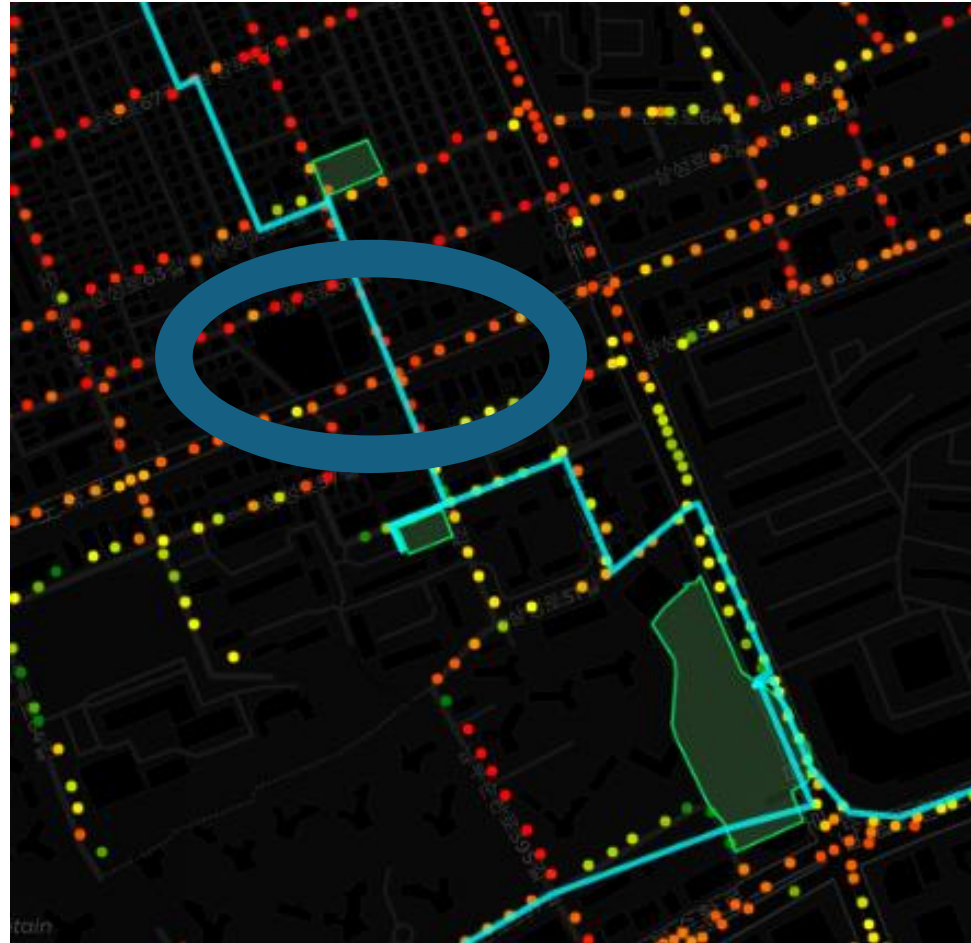


- Identify road segments with consistently low GVI values as **priority intervention zones**
- **Increase street greenery** in low-GVI areas
 - Additional street tree planting
 - Green buffers along sidewalks
 - Pocket parks and planter installation
 - Vertical greening on building facades

Result 3 – Example of Policy Recommendation



Result 3 – Example of Policy Recommendation



Limitation

1. Some **data are outdated** due to limited Google Street View coverage in Korea.
2. **Winter images** may reduce GVI accuracy.
3. **Traffic conditions** can affect GVI values.
4. Biodiversity requires **additional factors** beyond greenery, such as human disturbance and slope.

Conclusion

1. Gangnam's greenery is **fragmented**, and many streets still fall into **low-GVI categories**. The proposed street-view-based method can identify where greenery is weak and where corridor links are most feasible.
2. The main planning implication is to **prioritize low-GVI road segments** along the proposed green corridor with specific weight and combine street-tree planting, pocket parks, green buffers, and facade greening to **strengthen connectivity**.
3. The map provides a practical evidence base for turning isolated green patches into a more **continuous urban green corridor** network.

QnA